



Numerical Model for Mesange V2 Pre-Design

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Requirements

Structure		Aerodynamics/Propulsion	
Fairing	Von Karmann Geometry	Stability	- Fins geometry like Mesange V1 (Clipped Delta Fins). - Need to vary the geometrical properties to have a stability margin between 2 and 6
Recovery	No		
Avionics	Length: 350 mm Verify electromagnetic interferences between PCBs	Environment	Standard Atmospheric Model (Rocket-Py) // No wind
Upper Bay	He tank – Cylinder Ø210 mm with two semi-spherical bulkheads - Length : 350 mm for piping between He tanks and propellants tanks	Engine	- TORGOS Technology - Thrust : 3500 N - Mass Flow Rate : 1,62 kg/m^2 - Ejection Speed : 2160,50 m/s - ISP : 220 s
Tanks Bays	- Two separated cylinder Ø230 mm - Volumes variate	Propellants	- Depletion function is linear - Zero propellant mass after burn-out
Lower Bay	Idem to Mesange V1	Inertia Hypothesis	
Engine Bay	TORGOS	Cylinder	- Internal Components Bays - Tanks
Retrain	Study with and without	Hollow Cylinder	Fuselage Cylinder Part of the Rocket Body
Rocket diameter	Ø230 mm	Cone	Fairing
Launch-Ramp	12 m as for Mesange V1 - Inclination between [70°,85°]	Triangular Polygon	Fins
		Cuboid	Fins

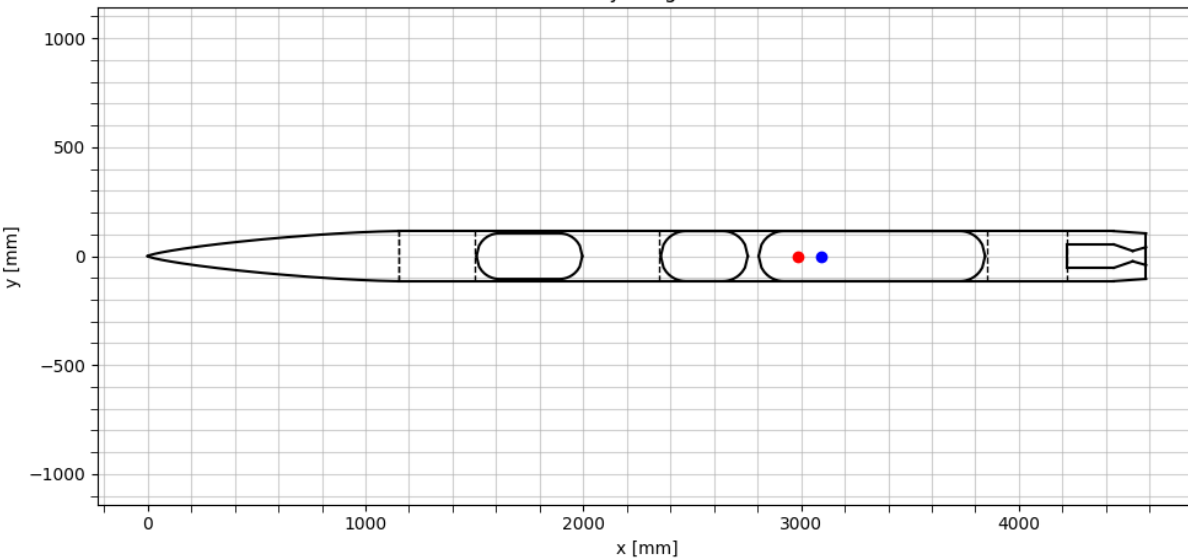
Work Methodology

Input DATA Parameters

- Payloads and Bays Masses
- TORGOS Engine Geometry and Performance Parameters
- Tanks Dimensional Parameters
- Rocket Fuselage Size and Design Parameters
- Bays Length Parameters
- Fins Design



Rocket Geometry Longitudinal Cut View



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-----Geometry Parameters - RASAero II-----
Payload Mass: 1 kg
Diameter Nosecone : 230 mm --- 9.0551 in
Length Nosecone : 1150 mm --- 45.2756 in
Avionics Bay Length: 350 mm --- 13.7795 in
Pressurization Bay Length: 845.75 mm--- 33.2973 in
Tank Bay Length : 1509.07 mm --- 59.4121 in
Lower Bay Length : 364 mm --- 14.3307 in
Upper Shell Length : 214.80 mm --- 8.4567 in
Boat tail Length: 150 mm --- 5.9055 in
Boat tail Rear Diameter: 209.02195641694686 mm --- 8.2292 in
Rocket Length: 4583.62 mm --- 180.4574 in
Cg before launch : 3093.94 mm --- 121.8088 in
Cg dry: 2985.83 mm --- 117.5524 in
Number of fins : 4
Root Chord : 364.0 mm --- 14.330709 in
Tip Chord : 190.0 mm --- 7.4803 in
Span : 228.6 mm --- 9.0000 in
Cant Angle : 0.00 deg
Sweep Distance :240.00 mm --- 9.4488 in
Peroxide Mass : 51.02 kg
Propane Mass : 6.80 kg
He Mass : 0.71 kg
Oxidizer and Fuel Mass : 57.82 kg
  
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Output plot use on

2

Work Methodology

RASAero 2

- Use the output plot to introduce the rocket design parameters into RASAero 2 software
- Extract the CD Power Off and CD Power On specific to the rocket design into a .csv

Methodology

- Copy/Paste Aerodynamical Results into an excel file
- Create 2 .csv documents with
- 1st Column (Mach) / 2nd Column (CD Power Off or On) and Delete the Others Column
- **ATTENTION ROCKETPY** only read .csv separated with , for columns and . for decimals
You can use (CD_Files.ipynb)

Thrust Curve

- Generate a thrust curve specific to TORGOS thrust and combustion time

Tool Used

Thrust_curve_generator
Generates a simplified curve

Stability

- Check CP position regarding the rocket center of mass
- Design the fin geometry regarding the previous statement

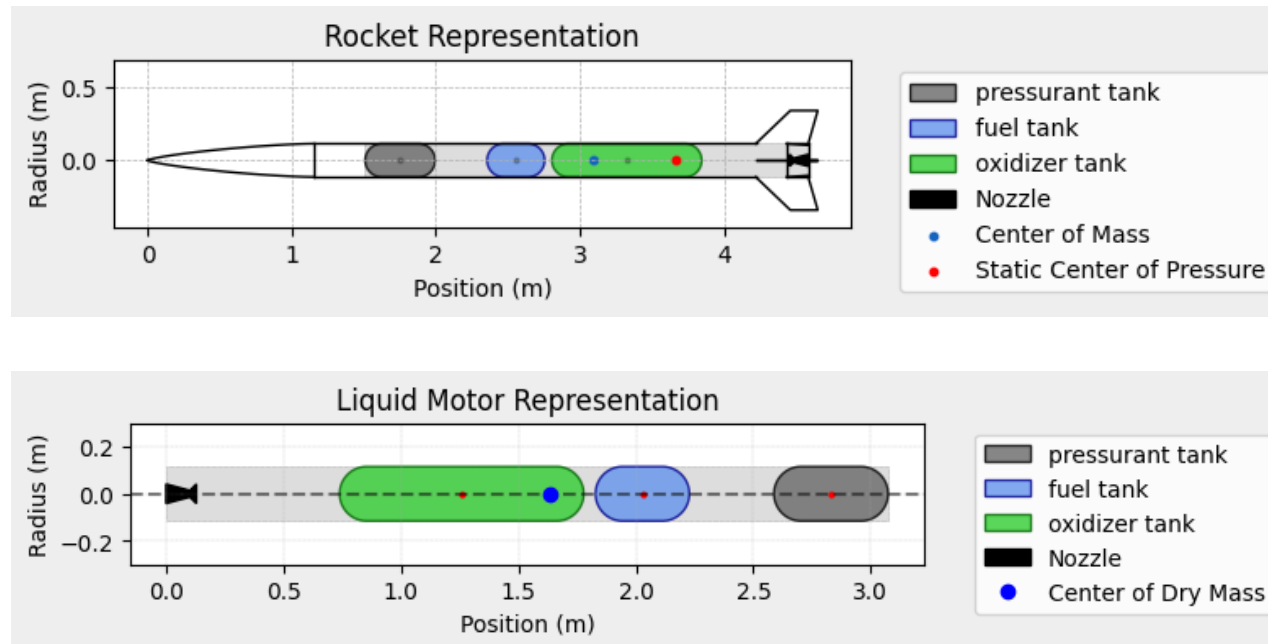
ATTENTION

DON'T FORGET TO EVERY TIME THE GEOMETRY IS CHANGED, THE FINS GEOMETRY ON THE PYTHON CODE SHOULD BE ALSO CHANGED

Work Methodology

Rocket-Py Simulation

- The user should not modify anything, as the program automatically considers the previous design parameters and the aerodynamical.csv file as input.
- Verify that the rocket's Liquid Motor Usage and overall, Rocket Usage reflect geometrical and fluidic behavior consistent with the previous data.



Recommendations for Optimization/Verification

- Verification of the Moment of Inertia Tensor and Center of Mass positioning method
- Replace Mass_Flow_Rate-Based_Tank for Ullage-Based_Tank for the propellant tanks to make them dependent on the pressure tank.
- Replace the Standard Atmosphere Model with one more realistic to constraint rocket trajectories.